EDS223 Homework 2

Exploring patterns of environmental justice

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Load libraries

```
library(sf)
library(tmap)
library(tidyverse)
library(spData)
library(gt)
```

Load data

```
# Load the data into respective variables
# EJ and census block data
ejscreen <- sf::st_read("data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb/")
```

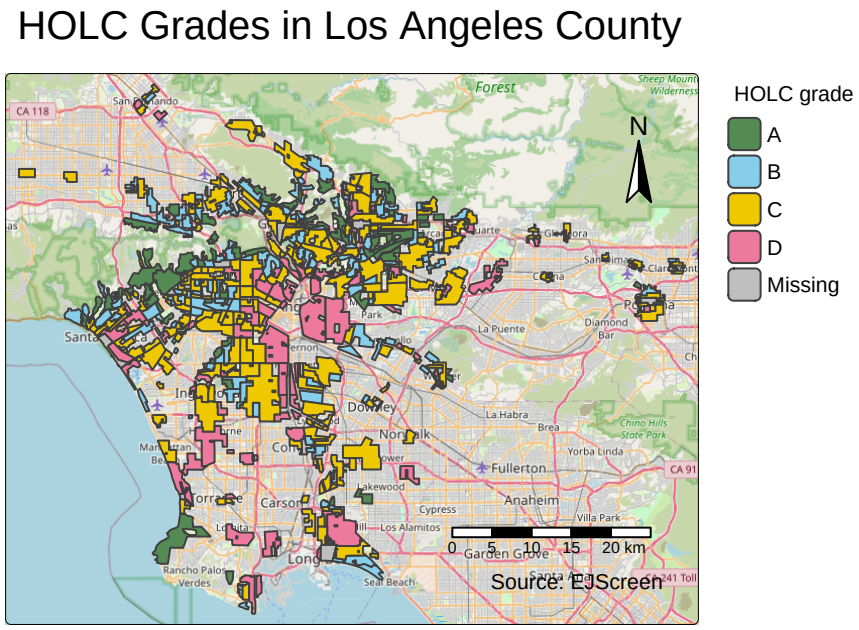
```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/edengonzalez/Desktop/MEDS/git/eds223-assignments/eds223-hw2/data/ejscreen/EJSCREEN_
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Crash Data
crash <- read_csv("data/SWITRS_LA/Crashes.csv")
# HOLC data
redlining <- read_sf("data/mapping-inequality/mapping-inequality-los-angeles.json")
```

Part 1: Legacy of redlining in current environmental (in)justice

Create a map of historical redlining neighborhoods

```
# Map of historical redlining neighborhoods
tm_basemap("OpenStreetMap") +  # Pick appropriate basemap
tm_shape(redlining) +          # Map the redlining data
  tm_polygons(fill = "grade",  # Fill neighborhoods colored by HOLC grade
    fill.scale = tm_scale(
      values = c("palegreen4", "skyblue", "gold2",
                 "palevioletred2", "white")),  # Reassign colors
    fill.legend = tm_legend(title = " HOLC grade",
    position = tm_pos_out("right"),
              frame = FALSE,
              title.size = 0.7)) +
  tm_scalebar() +  # Add scale bar
  tm_compass(position = c("right", "top")) +
  tm_credits("Source: EJScreen") +
  tm_title("HOLC Grades in Los Angeles County ") +
  tm_layout(frame.lwd = 0.4)
```



Create a table

Sumarize the percentage of census block groups that fall within each HOLC grade.

```
# Filter to keep values from Los Angeles County only
los_angeles <- ejscreen |>
  filter(CNTY_NAME == "Los Angeles County")

# Match redlining crs into los_angeles data frame's crs
redlining <- st_transform(redlining, st_crs(los_angeles))
```

```
# Verification that crs match
if(st_crs(los_angeles) == st_crs(redlining)){
  print("coordinate reference systems match!")
}else{
  warning("coordinate reference systems do NOT match")
  print(paste("los_angeles has ", st_crs(los_angeles)$epsg, "EPSG"))
  print(paste("redlining has",  st_crs(redlining)$epsg, "EPSG"))
}
```

```
[1] "coordinate reference systems match!"
```

```
# Use st_join to combine census block data with grade data
# This is now possible as they have been assigned the same geometries
la_red <- st_join(los_angeles, redlining) |>
  st_drop_geometry()  # No longer need the geometries
```

```
# Create the summary table with percentage of each percentage
# of census block groups that fall within each HOLC grade
summary_table <- la_red |>
  count(grade) |>  # Count the number of observations by grade
  # Divide by the total number of observations
  mutate(percentage = round((n / nrow(la_red)) * 100, digits = 2)) |>
  # Keep only the grade and percentage columns
  select(grade, percentage) |>
  gt() |>
  tab_header("Census Block Groups Per HOLC Grade")

# Print summary table
summary_table
```

Census Block Groups Per HOLC Grade

grade	percentage
A	5.00
B	13.79
C	34.02
D	14.98
NA	32.22

```
la_red |>
  count(grade)
```

```
  grade    n
1     A  449
2     B 1239
3     C 3058
4     D 1346
5  <NA> 2896
```

Create visualizations summarizing current conditions within HOLC grades

Wrangle Data

Use the mean for the analysis of percent low income, percentile of PM2.5, and low life expectancy.

```
# Mean percent low income per HOLC grade
mean_pct_low_inc <- la_red |>
  st_drop_geometry() |>
  summarize(mean_pct_low_inc = mean(LOWINCPCT, na.rm = TRUE), .by = grade) |>
  mutate(mean_pct_low_inc = mean_pct_low_inc * 100) |>
  drop_na()
```

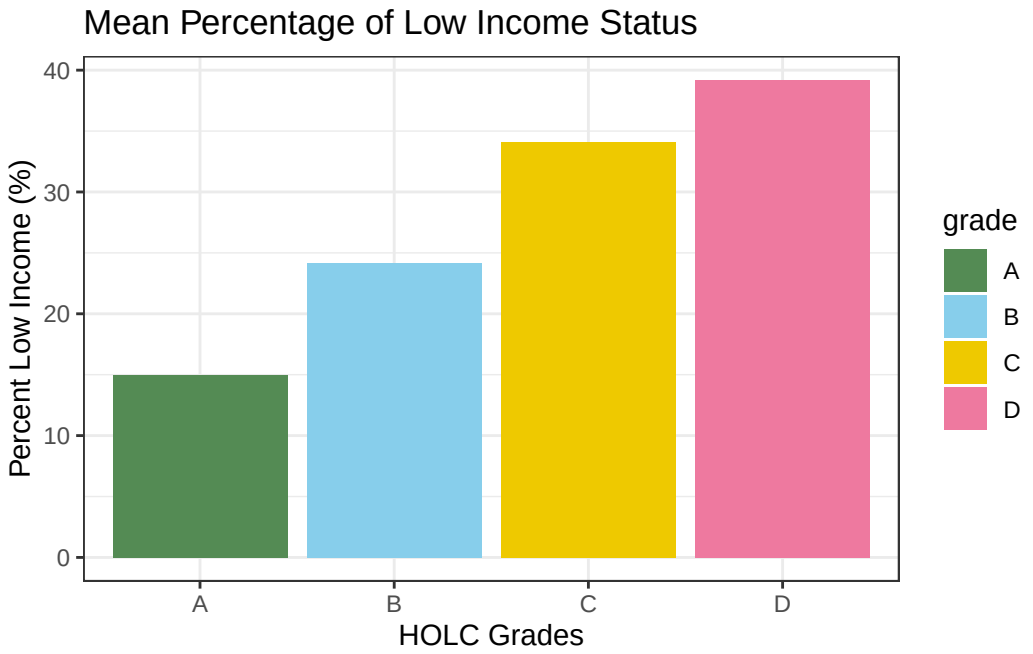
```
# Mean percentile of PM2.5 per HOLC grade
mean_pctl_pm25 <- la_red |>
  st_drop_geometry() |>
  summarize(mean_pctl_pm25 = mean(P_PM25, na.rm = TRUE), .by = grade) |>
  drop_na()
```

```r
# Mean percentile for low life expectancy per HOLC grade
mean_pctl_lif_exp <- la_red |>
  st_drop_geometry() |>
  summarize(mean_pctl_lif_exp = mean(P_LIFEEXPPCT, na.rm = TRUE), .by = grade) |>
  drop_na()
```

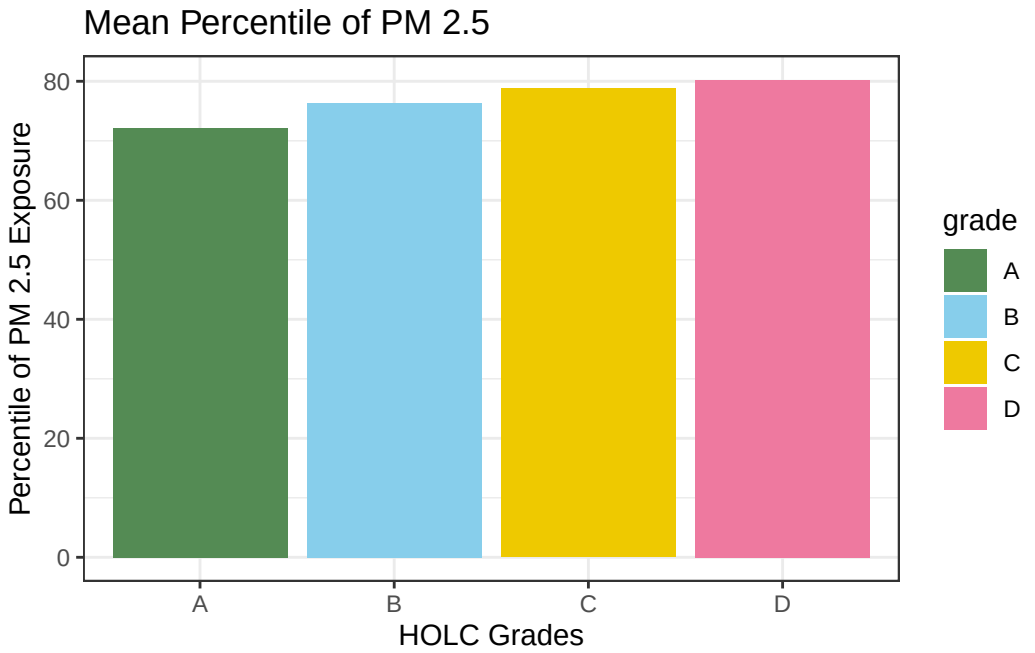
Plot the visualizations

```r
# Create a color dictionary to  assign colors for each grade
grades_colors <- c("A" = "palegreen4",
"B" =  "skyblue",
"C" = "gold2",
"D" = "palevioletred2")
```

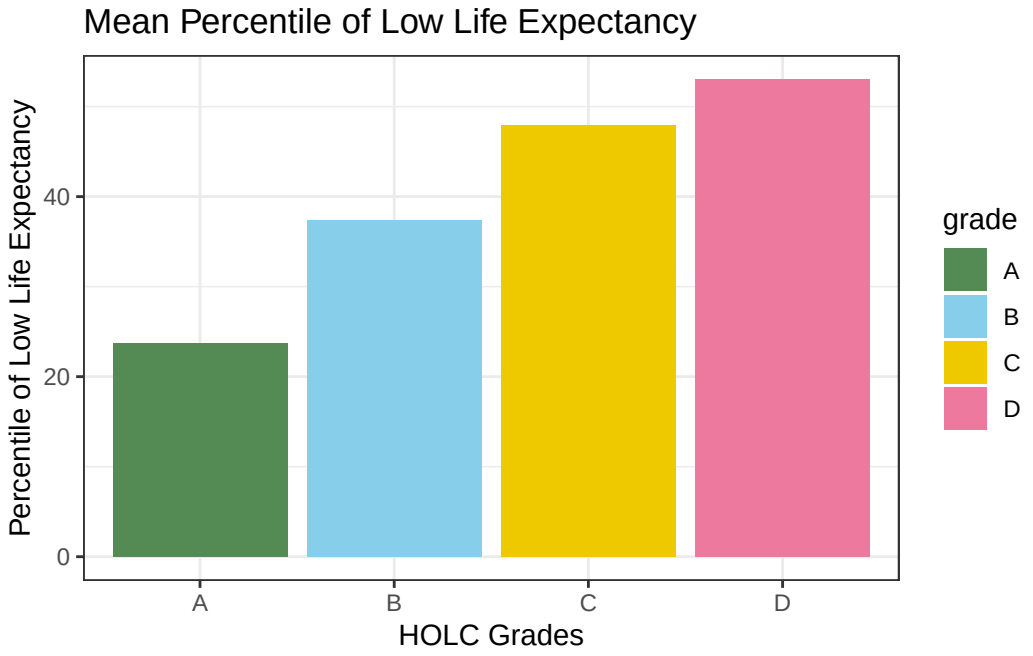
```r
# Plot the mean percent low income per HOLC grade
ggplot(mean_pct_low_inc, aes(x = grade, y = mean_pct_low_inc, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = grades_colors) +
  labs(x = "HOLC Grades", y = "Percent Low Income (%)",
  title = "Mean Percentage of Low Income Status") +
  theme_bw()
```



```r
# Plot the mean percentile of PM2.5 per HOLC grade
ggplot(mean_pctl_pm25, aes(x = grade, y = mean_pctl_pm25, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = grades_colors) +
  labs(x = "HOLC Grades", y = "Percentile of PM 2.5 Exposure",
  title = "Mean Percentile of PM 2.5") +
  theme_bw()
```




```r
# Plot the mean percentile of low life expectancy per HOLC grade
ggplot(mean_pctl_lif_exp, aes(x = grade, y = mean_pctl_lif_exp, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = grades_colors) +
  labs(x = "HOLC Grades", y = "Percentile of Low Life Expectancy",
  title = "Mean Percentile of Low Life Expectancy") +
  theme_bw()
```



Reflect

In all three graphs -mean percentage of low income status, mean percentile PM2.5, and mean percentile of low life expectancy- HOLC grade D had the highest values, C, B, A following in decreasing order. On average, about 40% of residents in census block groups within HOLC grade D are low income, compared to about 15% in grade A.
In general PM2.5 census block groups had high mean percentiles of PM2.5. Though, census block groups in grade D had higher mean than grade A (difference of 8.1 points on average). Lastly, on average, census block groups in grade D had higher low life expectancy (in the ~53.04 percentile) compared to census block groups in grade A (in the ~23.76 percentile), on average. In combination, the graphs highlight how the legacy of redlining persists as these neighborhoods are most burdened by poverty, air pollution, and lower life expectancy, on average.

Note: There are some logistical limitations in this analysis. There are some block groups that overlap two HOLC polygons and therefore get duplicated and counted in both grades.

Part 2: Legacy of redlining in pedstrian and bike accidents

```
# Bike and pedestrian acidents from crash data
bike_pedestrian <- crash |>
  # Filter for pedestrian and bicycle accidents
  filter(PEDESTRIAN_ACCIDENT == "Y"  | BICYCLE_ACCIDENT == "Y") |>
  # Drop values rows with empty coordinates
  drop_na(POINT_X, POINT_Y) |>
  # Assign the coordinates to make the data frame into a simple feature
  st_as_sf(coords = c("POINT_X", "POINT_Y"), crs = 4326) |>
  # Transform to match redlining CRS
  st_transform(st_crs(redlining))
```

```
# Check that coordinate reference systems match
if(st_crs(bike_pedestrian) == st_crs(redlining)){
  print("coordinate reference systems match!")
}else{
  warning("coordinate reference systems do NOT match")
  print(paste("bike_pedestrian has ", st_crs(bike_pedestrian)$epsg, "EPSG"))
  print(paste("redlining has",  st_crs(redlining)$epsg, "EPSG"))
}
```

```
[1] "coordinate reference systems match!"
```

```
# Keep the coordinates that intersect with the los angeles data
#  frame to make sure only coordinate points in la are mapped.
bike_pedestrian <- st_filter(bike_pedestrian, los_angeles)

# Use st_join to combine bike & pedestrian crash data with grade data
crash_red <- st_join(bike_pedestrian, redlining) |>
  st_drop_geometry()


# Check range of accident years
min(crash_red$ACCIDENT_YEAR)
```

```
[1] 2020
```

```
max(crash_red$ACCIDENT_YEAR)
```

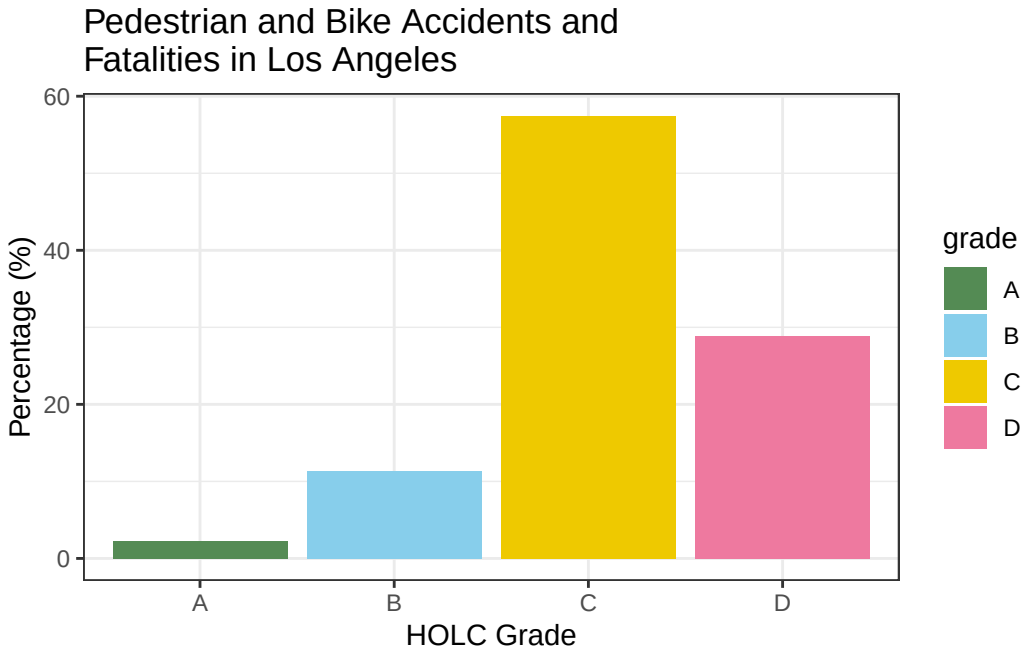
```
[1] 2025
```

```r
# Calculate percentage of pedestraian and bike crashes and
# fatalities per HOLC grade
crash_summary <- crash_red |>
  # Remove rows where grade is empty
  filter(!is.na(grade)) |>
  group_by(grade) |>
  summarise(n = n()) |>
  mutate(crashes = n / sum(n) * 100)

crash_summary
```

```
# A tibble: 4 x 3
  grade     n crashes
  <chr> <int>   <dbl>
1 A       360    2.29
2 B      1782   11.4
3 C      9026   57.5
4 D      4532   28.9
```

```r
# Graph the pedestraian and bike crashes and fatalities
# per HOLC grade
ggplot(crash_summary, aes(x = grade, y = crashes, fill = grade)) +
  geom_col() +
  scale_fill_manual(values = grades_colors) + # Assign the colors
  labs(x = "HOLC Grade",
       y = "Percentage (%)",
       title = "Pedestrian and Bike Accidents and\nFatalities in Los Angeles") +
  theme_bw()
```



The graph shows the percentage of pedestrian and bike accidents and fatalities in Los Angeles County by HOLC grade. When we drop the NA values, by far, census block groups of grade C had the highest percentage of pedestrian and bike accidents and fatalities (roughly 57.5%). HOLC grade D had the second highest percentage, roughly 28.9%. Meanwhile, areas with grade A had roughly 2.29% of pedestrian and bike accidents and fatalities in Los Angeles County.